ISYE4600 Project Proposal

Deliverable 1: Mid-Semester Project Proposal

1. Problem Statement

Autonomous vehicles and driver-assist systems are rapidly increasing in popularity and becoming more common in public roads. Even though these systems reduce human error, crashes still occur, and more often under complex road conditions. The teams behind these AVs need to decide where to prioritize safety improvements, whether that be in lane changes, disengagement handling or intersection behavior- they do this by looking through evidence from real crash reports. We will study those reported AV crash events and how factors in the context of the crash affect the severity of the crash, we will also look for recurring patterns in these crashes.

This is a high stakes problem for AV safety engineers and system designers because they need to identify which context and road conditions are most likely to produce severe outcomes- this allows them to prioritize efforts accordingly. This also affects regulators like NHTSA (National Highway Traffic Safety Administration) who monitor AV safety performance and use the reported crashes to make policy decisions.

It is currently hard to understand patterns and make predictive models because of the lack of structured analysis. Crash investigations are often case by case, making it hard to identify patterns across the different incidents. Other important risk factors can also be overlooked, as the dataset is high- dimensional and the crash narrative in each incident report is hard to analyze. This may lead to safety resources being allocated more inefficiently and in turn, to slower mitigation efforts.

2. Research Questions or Objectives

- Can we predict whether an AV crash will be severe (using injury severity, towing status, airbag deployment) based on crash context features?
- Can we identify recurring crash scenarios or failure modes using unsupervised clustering on a reduced set of meaningful crash context features?
- Can we predict whether a crash will be severe using interpretable supervised learning models such as logistic regression and random forest?
- Which contextual factors most strongly increase the probability of a severe crash?

Commented [VL1]: de pronto pondria estos depures de "features" por lo creo que son los actual features

Commented [VL2R1]: maybe I would put these after "features" because I think they are the actual features

3. Dataset Description

The dataset we will be using is from the National Highway Traffic Safety Administration (NHTSA) Standing General Order (SGO) on Crash Reporting for vehicles equipped with advanced driver-assistance systems (ADAS) and automated driving systems (ADS). The data is publicly available through NHTSA's website:

We will be using four datasets from the NHTSA Standing General Order crash reporting system:

- **Archived ADS Incident Reports (July 2021 – June 2025)**
  Archived_ADS_data
  - 2,295 incidents
  - 137 variables per record
  - Crashes involving Automated Driving Systems (Levels 3–5)

- **Archived Level 2 ADAS Incident Reports (July 2021 – June 2025)**
  Archived_level_2_ADAS_data
  - 4,029 incidents
  - 137 variables per record
  - Crashes involving Level 2 driver-assistance systems

- **ADS Incident Reports (June 16, 2025 – January 15, 2026)**
  Current_ADS_Incident_Report_data
  - 613 incidents
  - 116 variables per record
  - Automated Driving Systems (Levels 3–5)

- **Level 2 ADAS Incident Reports (June 16, 2025 – January 15, 2026)**
  Current_level_2_ADAS_Incident_Report_data
  - 849 incidents
  - 116 variables per record
  - Level 2 driver-assistance systems

Clarifications:

- Each row represents one reported crash event.
- Data includes vehicle info, roadway and weather conditions, automation status, crash context, severity indicators, and a short narrative text.
- Severity is defined using injury, towing, and airbag deployment.

Commented [VL3]: we said at the beginning that we will try to predict severity using those 3 features, we might need another metric that measures severity so that those 3 features can predict a metric whose value pertains only to a severity result?

Commented [VL4R3]: take into account that I haven't seen the datasets

- Level 2 systems assist the driver but require full human supervision.
- Levels 3–5 represent higher automation where the vehicle assumes primary driving responsibility.

Each row in the csv is a single AV crash report from NHTSA's Standing General Order dataset. It includes fields like vehicle info, roadway and weather conditions, automation status, crash context, and severity indicators, plus a narrative (unstructured). We treat each row as one incident and use injury, towing, and airbag deployment to define a severe vs non-severe outcome. Some incidents appear across multiple rows, we must account for these during the preprocessing of data by consolidating the records using the incident identifier.

4. Proposed Machine Learning Methods

We formulate this problem as a supervised binary classification task, where our goal is to predict whether an AV crash is severe or non-severe based on crash context features. We will define severity using injury status, airbag deployment, and towing.

We use logistic regression as a baseline model to provide a simple benchmark for severity prediction. This will allow us to examine model coefficients and understand how individual factors such as roadway type, automation engagement, and collision partner influence crash severity.

Commented [VL5]: ohhh ok now I understand… we give a severity score given injury, airbag, and towing and the features that we evaluate if they influence severity are these others that you mention

We compare this baseline against random forest and gradient boosting (XGBoost) models. Random forests capture nonlinear relationships and interactions between features while providing feature importance measures. We include Gradient boosting to improve the predictive performance by learning more complex patterns in the data and also addressing potential class imbalance.

We will also apply unsupervised clustering (K-means / hierarchical clustering) on a smaller set of crash context features, excluding variables that are metadata or noise. This will allow us to identify recurring crash scenarios.

These methods are well suited to our dataset and sample size:

Logistic regression provides transparency (binary), tree-based models capture nonlinear risk patterns, and clustering will aid in pattern discovery. The combination will give us a solid predictive capability and valuable safety insights.

5. Evaluation Plan

We will evaluate the supervised models using precision, recall, and F1 score due to the expected class imbalance between severe and non-severe crashes. These metrics will help us determine how well the models identify severe crashes while accounting for both false positives and false negatives.

To cover edge cases, we will also report confusion matrices to understand the classification behavior, and the rate of missed severe crashes. We use cross-validation to avoid relying on a single train test split and to get a more reliable estimate of the model performance. Logistic regression will serve as our baseline model, and we will evaluate improvements from random forest and gradient boosting relative to this benchmark.

For the unsupervised clustering analysis we will evaluate cluster quality using a silhouette score, this will tell us how well each data point fits within its assigned cluster. We will also examine dominant features within each cluster to determine whether the groups correspond to meaningful crash scenarios.

We define success by a measurable improvement in F1 score over the logistic regression baseline- and also by the ability to extract actionable safety insights, identifying high risk crash contexts (variables) and patterns associated with higher severity of crashes.

6. Action Plan

By Feb 23 - Data acquisition + data dictionary - Owner: Lauren

- Download 4 SGO datasets + document versions/time ranges
- Data dictionary + missingness summary
- Confirm incident identifier + duplication behavior

By Mar 2 - Cleaning / preprocessing - Owner: Camilo

- Merge/stack datasets + add indicators (ADS vs L2; archived vs current)
- Deduplicate/aggregate to one row per incident
- Build severity label + class distribution table

By Mar 9 - Baseline implementation - Owner: Santiago

- Majority-class baseline + simple logistic regression baseline
- Establish evaluation pipeline (split or CV, metrics: Precision/Recall/F1 + confusion matrix)
- Produce first performance table template

By Mar 30 - Model comparisons + tuning + clustering - Owners: Camilo & Lauren

- Train LR / RF / (XGBoost if allowed) + tuning on validation/CV
- Create performance tables + key figures
- Clustering (k-means/hierarchical) + silhouette + cluster profiles
- Start error analysis: examine false negatives / top failure contexts

By Apr 20 (draft) - Full report draft + reproducibility - Owners: Santiago & Lauren

- Draft report PDF with all required sections
- Add “Interpretation for the System” recommendations + limitations/next steps
- Write README with step-by-step reproduction instructions

- Organize code/notebooks + data download instructions

By Apr 27 (final) - Packaging + final edits - Owners: All

- Final proofreading + figure/table numbering
- Zip folder includes: PDF, code, data link/instructions, README
- Verify everything runs end-to-end from a clean environment